



BAULKHAM HILLS HIGH SCHOOL

2014

YEAR 12

TERM 2 ASSESSMENT TASK

Mathematics Extension 1

General Instructions

- Reading time – 3 minutes
- Working time – 65 minutes
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All relevant mathematical reasoning and/or calculations must be shown

Total marks – 36

This exam consists of 9 questions on pages 2-4.

Attempt all questions in the booklet provided.

36 marks

Attempt all questions

Answer each question in the answer booklet. Each answer sheet must show your BOS#. Extra Paper is available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 1

Sylvia knows that the equation $y = x^4 - 2x^2 + x - 3$ has a root between $x=1$ and $x=2$. She was asked to use the halving the interval method to estimate this root to the nearest integer. She correctly calculates $f(1) = -3$ and $f(2) = 7$. Sylvia then concluded that the root is closer to $x = 1$ than $x = 2$. Is Sylvia correct? Justify your answer. **2**

Question 2

Find $\int \frac{dx}{x + \sqrt{x}}$ using the substitution $u = \sqrt{x}$. **3**

Question 3

Evaluate the following definite integral using the substitution $u = \sin x$ **3**

$$\int_0^{\frac{\pi}{2}} \cos x \sqrt{\sin^5 x} \, dx$$

Question 4

It is known that $\log_e x + \sin x = 0$ has a root close to $x=0.5$. Use one application of Newton's method (taking $x=0.5$ as a first approximation) to find a better approximation to the root correct to two decimal places. **2**

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Question 5

(a) Sketch the graph of the function $y = \log_e(x - 2)$ 1

(b) The region bounded by $y = \log_e(x - 2)$, $x = 0$, $y = 0$ and $y = h$ is rotated about the y axis to create a bowl. 3

Find the exact volume of the bowl in terms of h .

(c) The bowl is placed with its axis vertical and water is poured into the bowl at the rate of 50 cm^3 per second. 2

Find the rate at which the water level is rising when the depth of water is 1.5 cm , giving your answer correct to 3 decimal places.

Question 6

A particle is moving on a straight line. At time t seconds it has displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. The particle starts from O and at time t seconds, $v = (1 - x)^2$.

(a) Find an expression for a in terms of x . 2

(b) Find an expression for x in terms of t . 2

(c) Find the time taken for the particle to slow down to a speed of 1% of its initial speed. 2

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Question 7

A cat sitting at the top of a wall $3.2m$ high sees a mouse on the ground $4m$ from the bottom of the wall. The cat jumps horizontally from the top of the wall with an initial velocity of $V \text{ ms}^{-1}$. Taking acceleration due to gravity to be 10 ms^{-2} and ignoring air resistance, the horizontal displacement equation of motion is:

$$x = Vt. \quad \text{DO NOT PROVE THIS}$$

- (a) Show that the vertical displacement equation of motion is $y = 3.2 - 5t^2$ 1
- (b) Find the time taken for the cat to reach the ground. 1
- (c) If the cat wants to land on the mouse calculate the speed at which the cat must jump horizontally from the wall. 1

Question 8

A particle moving in simple harmonic motion has its acceleration given by $\ddot{x} = -9x$ where x is the displacement in metres of the particle from the origin after t seconds.

- (a) Show that $x = -a \sin(3t + \alpha)$ is a possible equation of motion where $a > 0$ and α are constants and α is acute. 2
- (b) Initially, the particle's acceleration is 18 ms^{-2} and after $\frac{5\pi}{18}$ seconds the velocity is 12 ms^{-1} . Show that the amplitude is 4 metres. 2
- (c) Find the greatest speed and where it reaches this speed. 2
- (d) How many times does the particle change direction in the first 4 seconds? 2

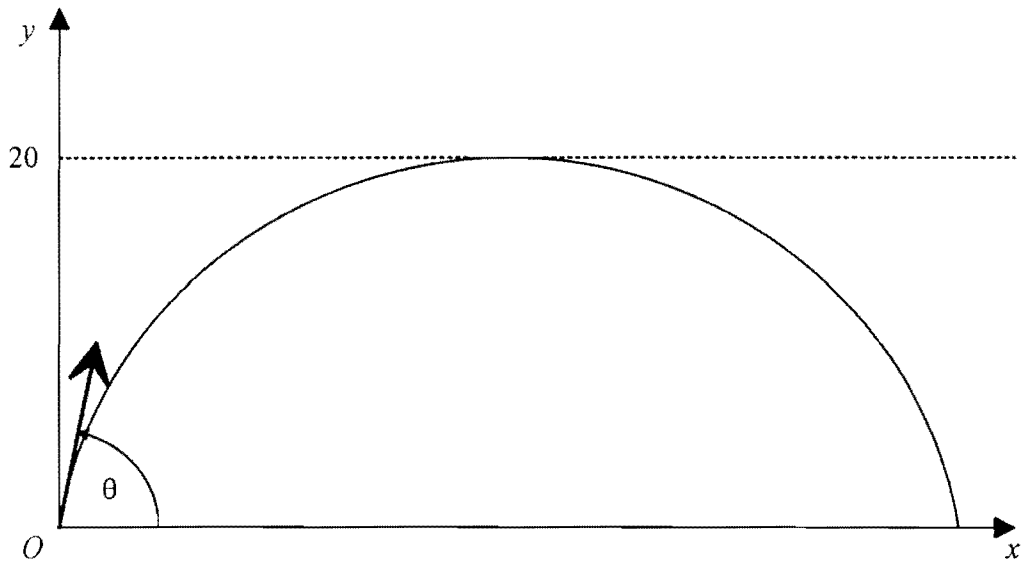
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Question 9

When a particle is fired in the open from a point O at a speed of 40 ms^{-1} and at an acute angle θ above the horizontal, the equations of motion are:

$$x = 40t \cos\theta \text{ and } y = 40t \sin\theta - 5t^2 \quad \text{DO NOT PROVE THESE}$$

The particle is fired with the same speed from a point O on the floor of a horizontal tunnel of height 20m .



Find the maximum horizontal range of the particle along the tunnel.

3

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2014 EXTENSION 1 HSC TASK JUNE SOLUTIONS

1. $f(1.5) = 1.5^4 - 2 \times 1.5^2 + 1.5 - 3$
 $= -0.9375$ ✓ calculates $f(1.5)$

∴ Root lies between $x = 1.5$ and $x = 2$

∴ Closer to 2 than 1. ✓

Sylvia is correct.

(Allow 1 mark if student answers 'not necessarily, she would need to find $f(1.5)$ ' without doing so.)

2. $\int \frac{dx}{\sqrt{x}(\sqrt{x}+1)}$ let $u = \sqrt{x}$
 $du = \frac{1}{2} x^{-1/2} dx$
 $du = \frac{dx}{2\sqrt{x}}$ ✓
 $= 2 \int \frac{dx}{2\sqrt{x}(\sqrt{x}+1)}$
 $= 2 \int \frac{du}{u+1}$ ✓
 $= 2 \ln(u+1) + C$
 $= 2 \ln(\sqrt{x}+1) + C$ ✓

3. $\int_0^{\pi/2} \cos x \sqrt{\sin x} dx$ let $u = \sin x$
 $du = \cos x dx$
 when $x = \pi/2$, $u = \sin \pi/2 = 1$
 $x = 0$, $u = \sin 0 = 0$ } ✓
 $= \int_0^1 u^{1/2} du$ ✓
 $= \frac{2}{3} \left[u^{3/2} \right]_0^1$
 $= \frac{2}{3} (1 - 0)$
 $= \frac{2}{3}$ ✓

4.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f(x) = \log x + \sin x$$

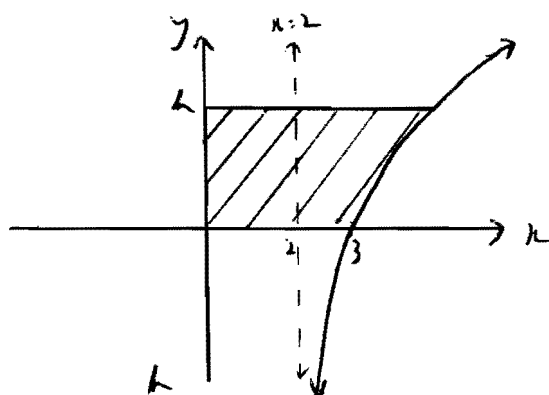
$$f'(x) = \frac{1}{x} + \cos x$$

$$x_1 = 0.5 - \frac{(\ln 0.5 + \sin 0.5)}{2 + \cos 0.5} \quad \checkmark$$

$$= 0.57427 \dots$$

$\therefore x_1 = 0.57$ (2dp) is a better approximation $\checkmark$

5. a)



b)

$$V = \pi \int_0^h x^2 dy$$

$$= \pi \int_0^h (e^y + 2)^2 dy \quad \checkmark$$

$$= \pi \int_0^h (e^{2y} + 4e^y + 4) dy$$

$$= \pi \left[\frac{e^{2y}}{2} + 4e^y + 4y \right]_0^h \quad \checkmark$$

$$= \pi \left(\left(\frac{e^{2h}}{2} + 4e^h + 4h \right) - \left(\frac{e^0}{2} + 4 + 0 \right) \right)$$

$$V = \pi \left(\frac{e^{2h}}{2} + 4e^h + 4h - \frac{9}{2} \right) \quad \checkmark$$

$$y = \log(x-2)$$

$$e^y = x-2$$

$$x = e^y + 2$$

c)

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$50 = \pi (e^{2h} + 4e^h + 4) \times \frac{dh}{dt} \quad \checkmark$$

When $h = 1.5 \text{ cm}$

$$\frac{dh}{dt} = \frac{50}{\pi(e^3 + 4e^{1.5} + 4)}$$

$$= 0.37882 \dots$$

$$\frac{dh}{dt} = 0.379 \text{ cm/s} \quad (3\text{dp}) \quad \checkmark$$

$$\begin{aligned}
 6 \ a) \quad v &= (1-x)^2 \\
 a &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 a &= \frac{d}{dx} \left(\frac{1}{2} (1-x)^4 \right) \quad \checkmark \\
 a &= 2(1-x)^3 \cdot -1 \\
 a &= -2(1-x)^3 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \frac{dx}{dt} &= (1-x)^2 \\
 \frac{dt}{dx} &= (1-x)^{-2} \\
 \int_0^t dt &= \int_0^x (1-x)^{-2} dx \\
 [t]_0^t &= \left[\frac{(1-x)^{-1}}{-1 \cdot -1} \right]_0^x \quad \checkmark \\
 t &= \left[\frac{1}{1-x} \right]_0^x \\
 t &= \frac{1}{1-x} - \frac{1}{1-0} \\
 t+1 &= \frac{1}{1-x} \\
 1-x &= \frac{1}{t+1} \\
 x &= 1 - \frac{1}{t+1} \quad \left(\text{or } x = \frac{t}{t+1} \right) \quad \checkmark
 \end{aligned}$$

$$c) \text{ Initial speed: when } x=0 \quad v = (1-0)^2 = 1 \text{ ms}^{-1}$$

$$1\% \text{ of initial speed, } v = 0.01$$

$$\begin{aligned}
 v &= \frac{dx}{dt} \\
 v &= (t+1)^{-2} \\
 v &= \frac{1}{(t+1)^2}
 \end{aligned}$$

$$\text{When } v=0.01 \quad 0.01 = \frac{1}{(t+1)^2}$$

$$(t+1)^2 = 100$$

$$t+1 = 10 \quad \therefore t = \underline{9 \text{ seconds}}$$

7 a)

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 dt$$

$$\dot{y} = -10t + C$$

when $t=0, \dot{y}=0$
 $0 = 0 + C$

$$C = 0$$

$$\therefore \dot{y} = -10t$$

$$y = \int -10t dt$$

$$y = -5t^2 + C_1$$

when $t=0, y=3.2$
 $3.2 = 0 + C_1$

$$C_1 = 3.2$$

$$\therefore y = 3.2 - 5t^2 \quad \checkmark$$

(1) correct solution

b) when $y=0$ $0 = 3.2 - 5t^2$
 $5t^2 = 3.2$
 $t^2 = 0.64$
 $t = 0.8 \quad \checkmark \quad (t > 0)$

c)

$$L = Vt$$

$$4 = V \times 0.8$$

$$V = 5$$

$\therefore$ Cat should jump at 5 ms^{-1} $\checkmark$

8 a) $x = -a \sin(3t + \alpha)$ (1)
 $\dot{x} = -3a \cos(3t + \alpha)$ ✓
 $\ddot{x} = 9a \sin(3t + \alpha)$
 $\ddot{x} = -9 \times -a \sin(3t + \alpha)$
 $\ddot{x} = -9x$ ✓ from (1)

b) when $t=0$, $\ddot{x} = 18$
 $18 = 9a \sin \alpha$

$$\sin \alpha = \frac{2}{a}$$

when $t = \frac{5\pi}{18}$, $v = 12$

$$12 = -3a \cos\left(\frac{5\pi}{18} + \alpha\right)$$

$$-4 = a \left(\cos \frac{5\pi}{18} \cos \alpha - \sin \frac{5\pi}{18} \sin \alpha \right)$$

$$-4 = a \left(\frac{-\sqrt{3}}{2} \cdot \frac{\sqrt{a^2-4}}{a} - \frac{1}{2} \cdot \frac{2}{a} \right)$$

$$-4 = \frac{-\sqrt{3} \sqrt{a^2-4} - 2}{2}$$

$$-8 = -\sqrt{3(a^2-4)} - 2$$

$$-6 = -\sqrt{3(a^2-4)}$$

$$36 = 3(a^2-4)$$

$$a^2 - 4 = 12$$

$$a^2 = 16$$

$$a = 4 \quad (a > 0)$$

c) max velocity at centre if in SHM i.e. $x=0$ ✓

$$\dot{x} = -12 \cos(3t + \alpha)$$

∴ max velocity is 12 ms^{-1} ✓ as $-1 \leq \cos(3t + \alpha) \leq 1$

d) $\sin \alpha = \frac{2}{4}$ from (b)

$$\therefore \alpha = 30^\circ$$

$$\therefore \dot{x} = -12 \cos\left(3t + \frac{\pi}{6}\right)$$

changes direction when $\dot{x} = 0$ since in SHM

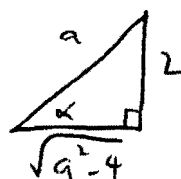
$$\therefore 0 = \cos\left(3t + \frac{\pi}{6}\right)$$

$$3t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}, \dots$$

$$3t = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \frac{10\pi}{3}, \frac{13\pi}{3}, \dots$$

$$t = \frac{\pi}{9}, \frac{4\pi}{9}, \frac{7\pi}{9}, \frac{10\pi}{9}, \frac{13\pi}{9}, \dots$$

∴ changes direction 4 times. ✓



8b) Alternative

Q8

If amplitude is 4m

$$x = -4 \sin(3t + \alpha)$$

$$\dot{x} = -12 \cos(3t + \alpha)$$

$$\ddot{x} = 36 \sin(3t + \alpha)$$

But $\ddot{x} = 18$ when $t = 0$

$$18 = 36 \sin \alpha$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = 30^\circ$$

check in $\ddot{x}$: when $t = \frac{5\pi}{18}$, $\ddot{x} = 12$

$$\ddot{x} = -12 \cos\left(\frac{5\pi}{6} + 30^\circ\right)$$

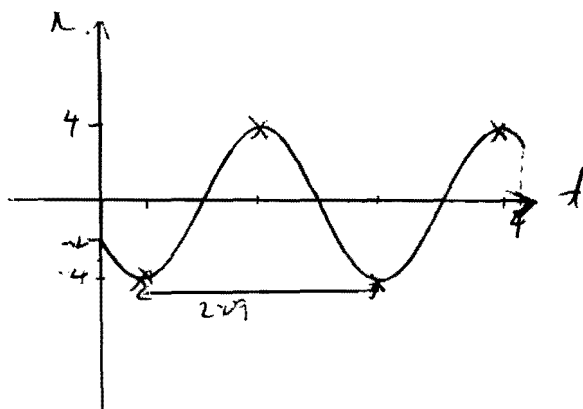
$$\ddot{x} = -12 \cos\left(\frac{5\pi}{6} + \frac{\pi}{6}\right)$$

$$\ddot{x} = -12 \cos \pi$$

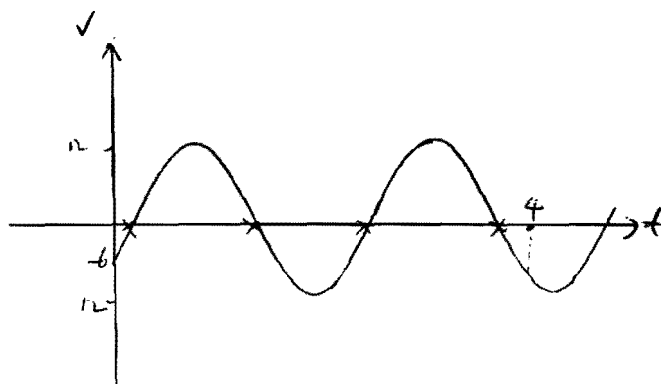
$\therefore \ddot{x} = 12$ as req'd.

$$\therefore a = 4.$$

8d) or graphically



4 changes



4 changes

9. Max range will occur if top of flight path is at a height of 20m.

$$y = 40t \sin \theta - 5t^2$$

$$\dot{y} = 40 \sin \theta - 10t$$

When $\dot{y} = 0$ max height is reached

$$0 = 40 \sin \theta - 10t$$

$$10t = 40 \sin \theta$$

$$t = 4 \sin \theta$$

✓

For max range $y = 20$.

$$20 = 10t(4 \sin \theta) - 5t^2$$

$$20 = 10t^2 - 5t^2$$

$$5t^2 = 20$$

$$t^2 = 4$$

$$t = 2 \quad \therefore \text{Time of flight} = 4 \text{ seconds}$$

✓

When $t = 2$, $2 = 4 \sin \theta$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\therefore \text{Max range} = 40 \times 4 \times \cos 30^\circ$$

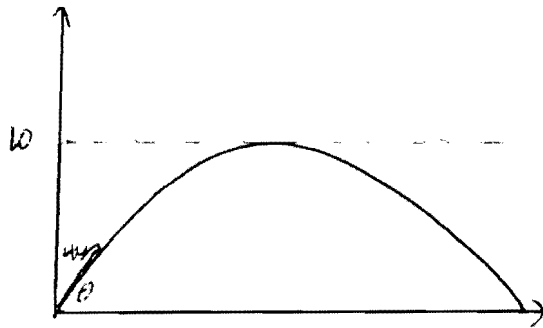
$$= 160 \cos 30^\circ$$

$$= 80\sqrt{3} \text{ m.}$$

✓

9.

method 2



$$x = 40 \tan \theta \quad y = 40 \sin \theta - 5t^2$$

$$\therefore t = \frac{x}{40 \cos \theta} \quad \text{sub in } y \rightarrow y = \frac{40 x \sin \theta}{40 \cos \theta} - \frac{5 x^2}{1600 \cos^2 \theta}$$

$$y = x \tan \theta - \frac{x^2}{320 \cos^2 \theta} \quad (i)$$

$$\text{when } y=0 \quad 0 = x \tan \theta - \frac{x^2}{320 \cos^2 \theta}$$

$$\frac{x^2}{320 \cos^2 \theta} - x \tan \theta + 10 = 0$$

for max range $0=0$

$$0 = \tan^2 \theta - \frac{4 \times 10}{160 \cos^2 \theta}$$

$$0 = \sin^2 \theta - \frac{1}{4}$$

$$\sin^2 \theta = \frac{1}{4}$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

for max range $y=0$ and $\theta=30^\circ$ in (i)

$$0 = x \tan 30^\circ - \frac{x^2}{160 \cos^2 30^\circ}$$

$$0 = \frac{x}{\sqrt{3}} - \frac{x^2}{320 \times \frac{3}{4}}$$

$$0 = x \left(\frac{1}{\sqrt{3}} - \frac{x}{240} \right)$$

starting point $x=0$ or

$$\frac{1}{\sqrt{3}} - \frac{x}{240} = 0$$

$$\frac{x}{240} = \frac{1}{\sqrt{3}}$$

$$x = \frac{240}{\sqrt{3}} \quad \text{or} \quad \frac{240\sqrt{3}}{3} = \underline{\underline{80\sqrt{3} \text{ m}}}$$